PRODUCT

CONFIDENTIAL

Technology Architecture

HyperClaw is built on PROMETHEUS intelligence layer, SOLOMON overmind, GENESIS self-expansion protocol, and four production-grade subsystems. Civilization Knowledge Base. Three Human Gates. 30+ platform connectors. 377 tests. MIT core. Zero compromises.

SYSTEM ARCHITECTURE — LAYERED STACK

PROMETHEUS

Intelligence layer. The recursive decision engine coordinating all agents and subsystems. Powers SOLOMON's orchestration decisions and GENESIS expansion logic.

Intelligence layer · Recursive orchestration

↑ SOLOMON (Overmind — always on, always watching) ↓

SOLOMON

Overmind agent. Always on, always watching. Routes tasks to 50+ domain specialists, triggers GENESIS when gaps are detected, enforces Three Human Gates.

Overmind · 50+ specialists · Three Human Gates

↕ GENESIS Protocol (self-expanding) ↕

GENESIS Protocol

Self-expanding: SOLOMON creates new specialist agents when it detects a capability gap. No other platform does this. Agents grow with the organization.

Self-expansion · Gap detection · Agent creation

↕

Ingests SOPs, job descriptions, org charts, client profiles, checklists,

Civilization Knowledge Base	runbooks, workflows, policies. Procedural chunker preserves step order. OrgGraph = role-aware context injection.	SOPs · OrgGraph · Procedural chunker · Interview agent

↕

HyperShield	Zero-trust security with hot-reload YAML policies. Three Human Gates: Ethical, Authorization, Uncertainty (<60% confidence). Full audit trail. Pre-execution enforcement.	Zero-trust · Hot-reload YAML · Three Human Gates

↕

UCB1 HyperRouter	Self-improving model routing. Learns optimal model-to-task assignment via UCB1 bandit algorithm. Formula: ucb1_score = mean_reward + C × √(ln(total_attempts) / attempts)	UCB1 bandit · Self-improving · Multi-model

↕

Three-Tier Memory	Session (fast, in-context) · Domain (persistent per-specialist) · Civilizational Graph (permanent, growing — pgvector). Causal relationships preserved across all tiers.	pgvector · Causal graph · Civilizational Graph

↕

HyperState	Typed, validated, immutable state management. Built on Pydantic v2. Every interaction serializable, auditable, and reproducible.	Pydantic v2 · Event sourcing · Supabase

↕

	30+ platform connectors: Telegram, Slack,	

HyperClaw Gateway	Discord, WhatsApp, Teams, Signal, Gmail, Google Workspace, MS 365, Notion, Salesforce, HubSpot, Jira, GitHub, Stripe, and more.	30+ connectors · Multi-platform · Enterprise integrations

↕

Infrastructure	Supabase (Postgres + pgvector + Auth), Python 3.11+, asyncio, OpenAI/Anthropic/Gemini SDKs, Redis (optional caching)	Supabase · Python 3.11+ · asyncio

CORE SUBSYSTEM DEEP DIVES

HYPERSTATE

Typed State Management

Every agent interaction is represented as a fully typed, Pydantic v2 validated state object. State transitions are atomic, logged, and replayable. No silent mutations — the system fails loudly if state integrity is violated.

```
class AgentState(BaseModel):
  task_id: UUID
  agent_id: str
  status: StateStatus
  context: dict[str, Any]
  created_at: datetime
  checksum: str
```

HYPERROUTER — UCB1

Intelligent Bandit Routing

UCB1 (Upper Confidence Bound) is a provably optimal exploration/exploitation algorithm from reinforcement learning. HyperRouter tracks success rates per agent per task type, automatically routing to the highest expected performer while still exploring alternatives.

```
UCB1 score = x̄ᵢ + √(2 ln(n) / nᵢ)

x̄ᵢ = avg reward for agent i
n = total tasks dispatched
nᵢ = tasks sent to agent i
```

HYPERMEMORY

Causal Graph Memory

Most vector memory systems store "what happened." HyperMemory stores why — causal edges between memory nodes let agents reason about dependencies, timelines, and consequences. Powered by pgvector for sub-millisecond semantic retrieval at scale.

HYPERSHIELD

Policy Engine

Before any agent executes an action, HyperShield evaluates it against a configurable policy ruleset. Rules can restrict data access, cap API spend, enforce content guidelines, or require human-in-the-loop approval. Full audit log for every decision.

```
Memory node schema:
{ id, embedding: vector(1536),
  content, timestamp, agent_id,
  causal_parents: [id...],
  causal_children: [id...] }
```

```
@shield.policy("no_pii_export")
async def validate(action: Action):
  if action.contains_pii():
    raise PolicyViolation(
      "PII export blocked")
```

PROMETHEUS + SOLOMON + GENESIS — INTELLIGENCE SYSTEM

⚡ PROMETHEUS Intelligence Layer

INTELLIGENCE LAYER · SOLOMON OVERMIND · GENESIS PROTOCOL · HYPERCLAW EXCLUSIVE

PROMETHEUS is HyperClaw's intelligence layer — the recursive decision engine that powers all high-level reasoning. It coordinates SOLOMON's orchestration decisions, drives GENESIS expansion logic, and manages multi-agent workflows with causal awareness.

SOLOMON is the overmind agent — always on, always watching. It monitors all 50+ domain specialists, routes tasks via UCB1 HyperRouter, enforces Three Human Gates (Ethical, Authorization, Uncertainty <60%), and triggers GENESIS when new capability gaps are detected.

GENESIS Protocol is HyperClaw's self-expansion engine. SOLOMON detects when no existing specialist can adequately handle a task — and automatically creates a new domain-specific agent to fill that gap. The agent network grows with the organization. No other platform has this capability.

Three Human Gates — baked into every workflow: (1) Ethical gate: flags potential policy violations before execution, (2) Authorization gate: role-based access control via OrgGraph, (3) Uncertainty gate: escalates to human review when confidence <60%. Enterprise-grade guardrails that compliance teams can actually approve.

QUALITY & TEST COVERAGE

377	0	MIT	v0.1.0
TESTS PASSING	KNOWN FAILURES	CORE LICENSE	ALPHA RELEASE

FULL TECHNOLOGY STACK

LAYER	TECHNOLOGY	PURPOSE	NOTES
Runtime	Python 3.11+	Core language	asyncio-native throughout

LAYER	TECHNOLOGY	PURPOSE	NOTES
State Validation	`Pydantic v2`	HyperState typing	Full type safety, JSON schema export
Database	`Supabase / Postgres`	Persistence layer	Row-level security, real-time subscriptions
Vector Memory	`pgvector`	HyperMemory embeddings	1536-dim vectors, HNSW index
LLM Providers	`OpenAI / Anthropic / Gemini`	Agent reasoning	Multi-model via HyperRouter
Caching	`Redis (optional)`	Response caching	Reduces LLM cost on repeated tasks
Testing	`pytest + asyncio`	377 passing tests	Unit + integration coverage
Packaging	`PyPI`	Distribution	pip install hyperclaw
Connectors	`HyperClaw Gateway`	Platform integrations	30+ connectors: Slack, Teams, Notion, GitHub, Stripe, and more
Source	`GitHub`	Open source core	github.com/mentatalbans/hyperclaw